

Truthful Peer Reports and Partial Identification of a Target Ranking

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Abstract

Peer scoring can strictly reward an observer for reporting a private comparison label, while the resulting report law leaves an intended Bradley–Terry order unresolved. We compose the quadratic peer score of Q with the four-vertex ambiguity in P: the same score table supports strict truthful reporting in two worlds with identical complete report laws and opposite endpoint orders. In a model with independent, positive, edge-specific shared flips, the strict comparisons identified by the truthful report law are exactly those given by directed reachability in the observed edge-sign graph. A conditional sign-margin bound gives a finite-sample version. These statements concern a restricted report-only game and a specified noise class; they neither challenge Q’s full-information implementation theorem nor select the truthful equilibrium.

1 Question and source boundary

The question is which intended Bradley–Terry comparisons are functions of a truthful-equilibrium peer-report law. Here Q means *Mechanism Design for Alignment and Control* [1], and P means the supplied local manuscript *When Peer Agreement Hides a Reversed Ranking* [4]. P’s own labels “Q” and “P” refer to a different pair of papers.

What Q establishes. For finitely many supported types, Q assumes that distinct types induce distinct, known conditional distributions $\beta_i[t]$ of co-player types. Against truthful co-player reports, its quadratic peer score gives type t a truthful-report advantage of $\Lambda \|\beta_i[t] - \beta_i[s]\|_2^2$ over a false report s . Its general implementation result also uses utility cancellation and an action-support penalty that may depend on the true state. Q permits correlated signals and gives the designer a model of state and types.

What P establishes. P models intended edge comparisons with Bradley–Terry score differences. On a four-vertex path, it constructs a clean world and a world with a common shared flip that have the same complete peer-report law but opposite intended endpoint orders. Under a separately assumed constant positive shared-flip mean, P also derives identification from a nondegenerate short cycle after observer reliability calibration.

2 A truthful equilibrium with an unidentified target

Orient the comparison path as $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$. For edge $e = (u, v)$, let $X_e \in \{-1, +1\}$ be an independent intended comparison with

$$\mathbb{E}X_e = \tanh\left(\frac{c_u - c_v}{2}\right).$$

In world W , the intended differences along the path are $(4, -3/2, -3/2)$. Each edge draw receives an independent common flip B_e of mean $b = 1/2$. In world W' , there is no common flip, and the intended differences are $f_b(d_e)$, where

$$f_b(d) = 2 \operatorname{artanh}(b \tanh(d/2)).$$

The path makes these transformed differences consistent with a Bradley–Terry score vector.

Two observers each receive a private label on every edge. Their reports are $Y_{ei} = X_e B_e Z_{ei}$ in W , with the common-flip factor omitted in W' . The personal flips Z_{ei} are independent across observers and edges, independent of comparisons, and have fixed means $a_i \in (0, 1)$ in both worlds. Observer i 's reportable type is the three-label vector $T_i = (Y_{ei})_{e \in E}$. Give each observer zero other utility and a singleton action set. The game thus concerns reporting an observed label vector.

Proposition 1 (Truthful reports do not identify the endpoint order). *In this two-observer game, one common quadratic peer-score table makes truthful label-vector reporting a strict best response for each observer in both worlds. The complete truthful-report laws coincide, yet $c_1 > c_4$ in W and $c'_1 < c'_4$ in W' . For any report-based binary endpoint-order decision, at least one world has error probability at least $1/2$, regardless of the number of independent repetitions.*

Proof. The shared latent sign $S_e = X_e B_e$ in W has mean

$$w_e = b \tanh(d_e/2) = \tanh(f_b(d_e)/2),$$

which is the mean of the intended sign in W' . These binary signs are independent over edges, and the personal-label channels agree. Hence the entire joint law of (T_1, T_2) , not merely its agreement moments, is the same in both worlds.

Let $\beta_i[t] = \mathcal{L}(T_j \mid T_i = t)$. On each edge, direct conditioning gives

$$\mathbb{E}[Y_{ej} \mid Y_{ei} = +1] - \mathbb{E}[Y_{ej} \mid Y_{ei} = -1] = \frac{2a_i a_j (1 - w_e^2)}{1 - a_i^2 w_e^2} > 0.$$

All label vectors have positive probability. Independence across edges therefore makes $\beta_i[t] \neq \beta_i[s]$ whenever $t \neq s$. Use Q 's report-only quadratic reward

$$R_i(r_i, r_j) = \Lambda \left(2\beta_i[r_i](r_j) - \|\beta_i[r_i]\|_2^2 \right), \quad \Lambda > 0.$$

When the other observer reports truthfully, misreporting s loses $\Lambda \|\beta_i[t] - \beta_i[s]\|_2^2 > 0$ in expectation. The identical joint label law gives the same score table in the two worlds.

The intended endpoint differences are

$$c_1 - c_4 = 1 > 0, \quad c'_1 - c'_4 = f_{1/2}(4) - 2f_{1/2}(3/2) \simeq -0.26458 < 0.$$

Any decision based on the reports has the same distribution in both worlds. Its two error probabilities sum to one, so their maximum is at least $1/2$. $\square$

The proposition asserts existence of a strict truthful equilibrium in the report-only specialisation. It does not assert that this equilibrium is unique. The label vectors also need not be identical full universal types in Q 's model: those types include beliefs about the state, and Q gives the designer a state-and-type model.

3 Which orders survive edge-specific shared flips?

Now retain independent edge draws and personal-flip channels, but permit each edge to have an unknown positive shared-flip mean $b_e \in (0, 1]$. For an arbitrarily oriented edge $e = (u, v)$, its calibrated corrupted-sign mean is

$$w_e = b_e \tanh\left(\frac{c_u - c_v}{2}\right).$$

Assume the population means are feasible under this model and nonzero. The sign of w_e is available either after calibration or directly from a truthful observer whose reliability is known to be positive. Orient each edge from its higher-score endpoint to its lower-score endpoint according to that sign, producing a directed graph H .

Proposition 2 (Exact population ordinal certificate). *Within this positive edge-specific flip model, the strict inequality $c_u > c_v$ is forced by the complete truthful-report law exactly when H has a directed path from u to v . If neither vertex reaches the other, two admissible worlds have the same complete report law and opposite strict orders for that pair.*

Proof. Each directed edge has a positive score drop, so a directed path forces its endpoint inequality. Feasibility makes H acyclic. If u and v are incomparable, choose a linear extension with u above v and another with their positions reversed. In each extension, assign descending scores with adjacent gaps exceeding $\max_e 2 \operatorname{artanh} |w_e|$. For each edge put

$$b_e = \frac{w_e}{\tanh((c_u - c_v)/2)} \in (0, 1).$$

Both score assignments give the prescribed sign and mean w_e on every corrupted binary edge sign. Independent edge draws and identical personal-flip channels therefore yield the same complete peer-report law, while the chosen pair reverses. $\square$

If a calibrated w_e is zero and $b_e > 0$, its endpoints have equal scores. Such edges may be contracted when the resulting data remain feasible; the strict-order statement then applies to the quotient. Under the stated class, a complete comparison graph is necessary and sufficient for a uniform certificate of every pairwise order across all distinct-score configurations. This is a uniform graph statement, whereas the proposition is specific to the realised signs.

The restriction on shared-flip means matters. P's constant-rate, nondegenerate-triangle formula can identify the common rate and repair its four-vertex path after adding edge 1–3, provided observer reliabilities are calibrated. The formula cannot verify rate constancy from reports: the recorded triangle-plus-tail construction has the same report law with positive edge-specific rates and opposite 1-versus-4 orders. A direct 1–4 comparison does certify that ordinal order under positive edge-specific flips.

Proposition 3 (Conditional finite-sample sign certificate). *Suppose truthful reports provide N independent labels per edge from one observer with positive reliability at least $\alpha > 0$ on every edge. If $|w_e| \geq \kappa > 0$ for every edge, then*

$$N \geq \frac{2}{\alpha^2 \kappa^2} \log\left(\frac{2|E|}{\zeta}\right)$$

is sufficient for all empirical edge signs, and thus every order certified by directed reachability, to be correct with probability at least $1 - \zeta$.

Proof. Each label has mean $a_e w_e$, with absolute value at least $\alpha \kappa$. Hoeffding's inequality gives sign-error probability at most $\exp(-N \alpha^2 \kappa^2 / 2)$ for one edge. A union bound over edges yields the stated sufficient sample count. $\square$

4 Related work and interpretation

Miller, Resnick and Zeckhauser [3] already use peer reports to reward truthful signals; Shnayder and co-authors [5] study uninformative equilibria. Chen, Feng and Yu [2] already obtain strict truthful elicitation of pairwise comparisons under Bayesian strong stochastic transitivity. Our claim is restricted to the composition with P's report-law reversal and the positive edge-specific flip criterion. The reachability step is elementary, and the searches in `search.md` do not establish priority for this precise formulation.

Strategic truthfulness leaves the intended target outside the report law in the two-world example. Within the positive edge-specific flip class, a directed chain certifies its endpoint order; an incomparable pair requires further information, such as a direct comparison. This interpretation is conditional on the stated model and says nothing about downstream decision payoffs.

5 Limitations and open questions

The first proposition has label types, a singleton action and zero other utility. Q's broader theorem assumes known beliefs and may use state-contingent rewards. We establish neither a refutation of that theorem, equilibrium selection, nor truthfulness with a score table estimated from strategic reports.

The graph criterion assumes independent edges and positive shared-flip and observer means. Unknown or negative reliability signs remove its guarantee. The sample bound assumes truthful reports and a positive margin; without that margin there is no uniform rate. Finite-sample cycle calibration remains open.

Open work is to justify shared-noise restrictions externally, analyse strategic score-table estimation and equilibrium selection, and connect an identified comparison to a decision payoff and label cost.

References

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